

LOADING BAY EQUIPMENT

GANDHI
Automations Pvt Ltd

India's No.1 Entrance Automation & Loading Bay Equipment Company

EUROPEAN COLLABORATION AND TECHNOLOGY WITH INNOVATIVE AND CREATIVE ENGINEERING FOR MORE THAN 20 YEARS





SAFETY AND QUALITY
GO HAND IN HAND

A PARTNER WHO STANDS BY YOUR SIDE

Quality & Safety - Our Priority

The Dock Levelers and Sectional Overhead Doors are CE certified, manufactured in accordance with

STANDARD
EN 1398
EN 13241-1:2003+A1:2011

DIRECTIVES : 2006/42/CE
2004/108/CE

Gandhi Automations develops and manufactures its products strictly in line with these standards. You are continuously assured safe operation with regular use, assembly and maintenance.

10 YEARS
GUARANTEED AVAILABILITY

CRISIL RATED COMPANY

Indian Green Building Council
M E M B E R

TÜV NORD ISO 9001
An ISO 9001 : 2008 Company

CE

ORIGINAL SPARE PARTS
Guaranteed original spare parts availability for 10 years

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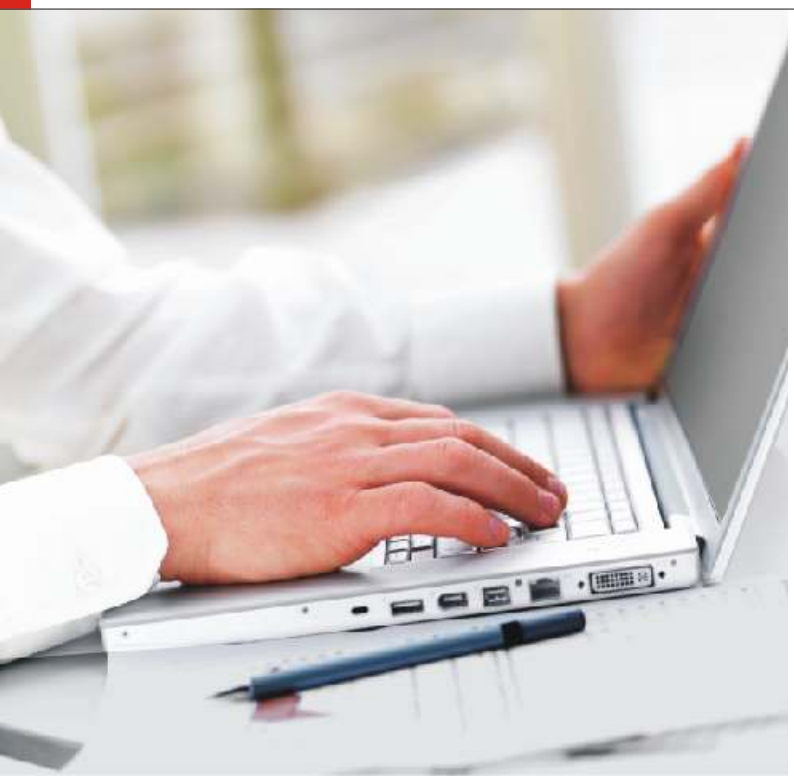
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Our Quality For Your Safety





Gandhi Automations Pvt Ltd is India's No.1 Entrance Automation and Loading Bay Equipment Company. This widely recognized position has been achieved over years of hard work, innovation, commitment to quality and reliable customer service. Since 1996 we have been designing, manufacturing, exporting and installing products that are problem free, safe and easy to operate.

Head Office

Headquartered in Mumbai; the commercial capital of India, the company has expanded its manufacturing capacity, operations, geographical reach and customer base globally. Today our products are marketed by us and our distributors in more than 15 countries across Asia, Europe, Africa and Americas.

Competent Advice

Proper customer guidance and service are our top priorities. Our well trained and experienced specialists are always at your disposal from the project planning stage to final building inspection. They help you to choose a perfect product solution tailored to your requirements.

Quality Management

The company is certified to ISO 9001 : 2008 by TÜV Nord. This has resulted in the implementation of continuous improvement in personnel training, production, equipment calibration, machinery maintenance, logistics and customer relations.

Nationwide Network

- Ahmedabad
- Bengaluru
- Bhopal
- Bhubaneswar
- Chandigarh
- Chennai
- Coimbatore
- Goa
- Hyderabad
- Indore
- Jaipur
- Kochi
- Kolkata
- Lucknow
- Mangalore
- Nagpur
- New Delhi
- Pune
- Raipur
- Ranchi
- Surat
- Vadodara
- Visakhapatnam





Advance Manufacturing

Our strategically located state of the art manufacturing and warehousing facility of 3,00,000 sq ft near Bhiwandi (40km N-E of Mumbai) ensures high quality products and short delivery times.

Inhouse research and development is an important part of the Gandhi Automations philosophy.

All essential product components and control technology are designed and produced by Gandhi Automations with European collaboration and technology.

Our product engineering team uses latest software combined with technologically advance machinery to offer the customer an excellent engineered product.

Packaging

Special care is taken while packaging our products for dispatch to the installation sites. Each consignment is supervised by a team of experts who ensure that material is dispatched efficiently and promptly.

Install, Commission & Service

Our team of committed and diligent engineers and technicians; many of whom are trained at advance training centers in Europe ensure perfect installation and trouble free commissioning according to our specifications assuring product reliability and life.

Dynamic Customer Service



Our customer care team offers 24x7 service. We lay great emphasis on exceptional aftersales service and provide regular programmed inspections and maintenance safeguards which increases product lifetime and reduces operational cost.

Our widespread service network assures response time to a minimum.

In rare case of any product failure, all you need to do is reach for the telephone.

TOLL FREE
1800 209 0200
From Anywhere in India

One call and one of our highly trained service technicians will be on its way in a fully equipped customer service van with wide range of original spares.

Loading Bay Equipment - Quality Engineered





Gandhi Automations offer complete logistics solutions by providing Dock Levelers, Dock Shelters, Sectional Overhead Doors and Dock Houses. As all components for your Loading Bay are available from a single source, the whole process - from product development and design to manufacture, installation and service are optimally co-ordinated, which ensures smooth loading and unloading at your Loading Bay.

The Dock Levelers from Gandhi Automations are manufactured conforming to the European Standard EN 1398 and using the advanced robotic welding technology to create a precision engineered product.

Dock Levelers allow quick and safe loading and unloading of goods from the vehicles thus minimizing energy used, saving on heating and chilling costs, resulting in maintaining the quality of the transported goods.

Dock Shelters provide an optimum seal between the internal and external environments and assist in reducing the energy consumption. Dock Shelters in combination with a Dock Leveler eliminates the need for a ramp and canopies.

Sectional Overhead Doors are ideal for closing a loading bay; insulated Rolling Shutters can also be utilized alternately.

Gandhi fabricated Dock Houses form a complete self-contained loading bay installed directly on the building thus maximizing usable space within the building.

Dock Levelers



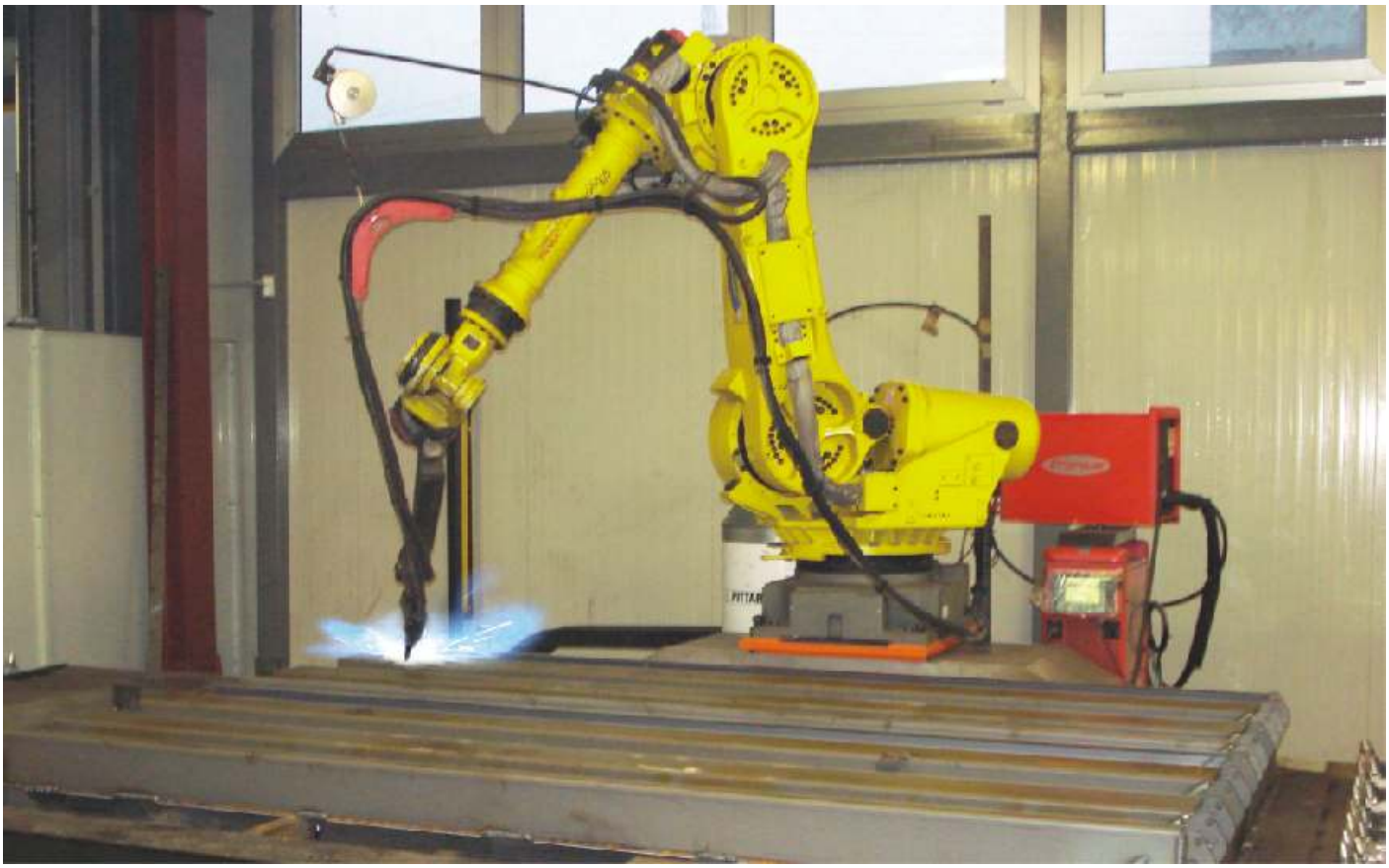
Dock Levelers offered by Gandhi Automations are designed as per EN 1398 standard for the most demanding loading and unloading operations. Dock Levelers facilitate smooth and safe transition by compensating the difference in heights between the loading bay and the vehicle. A Dock Leveler is not simply a bridge for connecting a vehicle but guarantee an efficient working environment to comply with the safety in work regulations.

The platform is constructed for a carrying capacity of 6t or 9t point load in almond anti-slip steel, with degreasing, phosphating, oven dried epoxy antirust + epoxy final coat finish with robotically welded supporting beams to create a precision engineered product. Self-supporting base construction.

The sturdy articulated steel construction is safely and reliably controlled via an electro-hydraulic system. A single push button activates the Dock Leveler into operation.

The most advanced concept of our Dock Leveler is to have the whole drive unit contained in a wall box which is installed on a wall inside the warehouse, at eye level, which allows for easy and economical maintenance, without the necessity to manoeuvre under the platform or inside the pit, where traditional powerpacks and controls are usually installed: an undoubtedly dangerous, dirty and uncomfortable operation.

All Dock Leveler controls offer the option of mutual safeguarding of the Dock Door and Dock Leveler to prevent damage.



It goes without saying that our Dock Levelers meet all the relevant standards and directives:

- EN 1398 Dock Levelers
- Machine Directive 2006/42/CE
- Low-Voltage Directive 2006/95/CE
- Electromagnetic Compatibility Directive 2004/108/CE

Radius Lip Dock Levelers



Radius Lip Dock Levelers allow the dock to connect with the truck bed, thus making it possible to drive directly on and off with fork-lift trucks etc. The self-cleaning lip hinging system does not retain rubbish with automatic end-of-run, so as to keep the 25 mm security distance between the folded lip and structure as per EN1398 & EN 349.

*Standard dimensions in mm of the BASE types, 6t point load (15t evenly distributed) or MAXI 9t point load (22t evenly distributed).

MODEL	PLATFORM WIDTH	PLATFORM LENGTH	LIP LENGTH	PIT WIDTH	PIT HEIGHT
RDL 2025	2000	2500	410	2040	550
RDL 2030	2000	3000	410	2040	550
RDL 2035	2000	3500	410	2040	550
RDL 2040	2000	4000	410	2040	550
RDL 2125	2100	2500	410	2140	550
RDL 2130	2100	3000	410	2140	550
RDL 2135	2100	3500	410	2140	550
RDL 2140	2100	4000	410	2140	550
RDL 2225	2200	2500	410	2240	550
RDL 2230	2200	3000	410	2240	550
RDL 2235	2200	3500	410	2240	550
RDL 2240	2200	4000	410	2240	550

*Other dimensions available on request.

Telescopic Lip Dock Levelers



Telescopic Lip Dock Levelers are ideal for connecting vehicles unable to drive near dock i.e. sea containers, side loading railway wagons etc. These types can be supplied with a lip extending up to 1 m.

*Standard dimensions in mm of the BASE types, 6t point load (15t evenly distributed).

MODEL	PLATFORM WIDTH	PLATFORM LENGTH	LIP LENGTH	PIT WIDTH	PIT HEIGHT
TDL 2025	2000	2500	500 - 1000	2040	550
TDL 2030	2000	3000	500 - 1000	2040	550
TDL 2035	2000	3500	500 - 1000	2040	550
TDL 2040	2000	4000	500 - 1000	2040	550
TDL 2125	2100	2500	500 - 1000	2140	550
TDL 2130	2100	3000	500 - 1000	2140	550
TDL 2135	2100	3500	500 - 1000	2140	550
TDL 2140	2100	4000	500 - 1000	2140	550
TDL 2225	2200	2500	500 - 1000	2240	550
TDL 2230	2200	3000	500 - 1000	2240	550
TDL 2235	2200	3500	500 - 1000	2240	550
TDL 2240	2200	4000	500 - 1000	2240	550

*The dimensions are nominal only. The technical drawings, supplied on demand, are to be considered valid for installation and operation.

All Hydraulic Dock Levelers in compliance to EN 1398



All tools pertaining to the Machine Directive 2006/42/CE are supplied with conformity certification and mark CE

Compulsory Norms for the Dock Levelers

- When choosing the dimensions of the Dock Leveler, the prescriptions of the EN 1398 are to be considered. It demands a width as wide as the truckbed, and a minimum of 700 mm wider than the forklift truck. The most common choices are widths of 2200 mm side flaps reducing the width for smaller trucks.
- The length has to form a maximum slope affordable for the forklift trucks (max. 8%) or electric pallet trucks (max. 3-4%), thus a standard length of 2500, 3000 mm is usual. The maximum slope of 12.5% stated by the EN 1398 is only the “slippery” limit.
- Standard carrying capacity of the dock leveler: uniformly distributed load of 15t / 22t (or point load of 6t / 9t as defined by EN 1398 totally concentrated on one axle of forklift truck, divided on two wheel prints of 150 x 150 mm at 1m distance).
- Anti-tripping yellow-black stripes 50 mm high under the rim of the leveler and of the pit, for the whole length
- Crossing of the leveler in rest position with full capacity
- Disjointable electric feeding as per EN 60204-1

Important Characteristics of the Dock Levelers

- CE European Certification. Built in conformity with EN 1398.
- Robotically welded platform and lip in anti-slip steel.
- Surface: Degreasing, Phosphating, oven dried epoxy antirust + epoxy final coat.
  Standard colour Dock Leveler : Grey RAL 7005
  Radius Lip : Orange RAL 2000
  Telescopic Lip : Grey RAL 7005
  Hot deep galvanising and non standard colours available on request.
- Electro hydraulic Dock Leveler allows connection to both positive and negative level differences. Maximum positive excursion: 640mm; Maximum negative excursion: 350mm
- Safety stop valve integrated in the lift cylinder: it blocks the Dock Leveler in case of accidental departure of the vehicle.
- Automatic side flaps for the reduction of the width of the lip, for narrow truck beds, automatically closing when the lip is extended and touches the sides of the van.
- Connection between dock and leveler without cutting points of the forklift wheels.
- Lift cylinder- Single effect; Lip cylinder-Double effect, self-venting.
- For swivelling lip levelers the self-cleaning lip that reduces the normally present dock debris, pallet pieces, nails, screws and bolts etc.
- Maximum pressure valve.
- Galvanised steel sheet foot side protections prevent feet from getting trapped between the loading platform and the Dock Leveler.
- Yellow / black side strips to evidence possible steps when the Dock Leveler is in use or underneath the dock.
- Maintenance crutch with positive engagement serves to secure and support the platform during maintenance checks and repairs.
- Lip support in resting position to prevent any cutting or overwhelm.
- Various sizes of rubber bumpers available to stop the vehicle and prevent damage to the Leveler.
- Tilt allows Dock Leveler to compensate for canted condition.
- Operator & control Unit: Electrohydraulic operator 400 V 50 Hz IP 55. Membrane Push Button with press & hold actuation and emergency stop integrated in IP 65 enclosure.



Dock Levelers

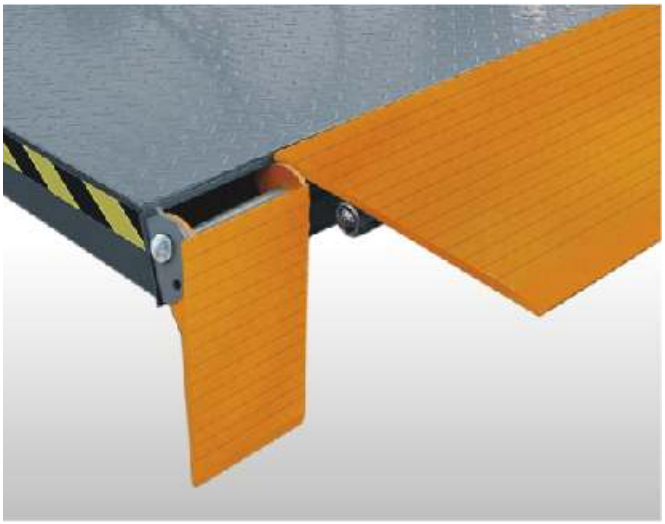




Dock Leveler is pre-mounted on a frame, to be laid on a slab built on the bottom of the traditional pit. The leveler is welded at the head and foot of the pit onto the profiles and is powered by electro hydraulic powerpack inside the pit.

Side Flaps

Automatic side flaps allow the choice of a wider platform Dock Leveler in conformity with the EN 1398 whilst maintaining the ability to level narrow vehicles.





Ready for pouring

Prefabricated Pit

The prefabricated pit or dock tray allows a perfectly squared, flush with pavement pit which is aesthetically pleasing and represents the complete solution to all problems of building a pit. It is assembled in less than a quarter an hour without any possibility of mistakes. The strong steel sheets duly bracketed are structurally calculated to substitute the three side reinforced walls. The monolithic front connection with bumpers hanger steel supports is a guarantee of resistance against the shocks of the docking vehicles. It is not necessary to build the perimetrical reinforced walls, which reduce adherence. No reinforcement is necessary. The risk of out-of-squaring pits is eliminated, thanks to the squaring jig. Also available is pit with clearance to receive tail-lift.

Accessories for Safe Docking





Lorry Wheel Guides for Perfect Docking

Wheel guides are hot dip galvanised steel guides consisting of two round shaped pipes 2 metres long and appropriate diameter. Fixed on the court pavement, in front of the loading bay they grant a perfect docking of the approaching vehicles (10 cm tolerance) and preservation of the equipment. Wheel guides do not affect the wheels of the lorry and does not reach their rims.

Bumpers

Bumpers are used to indicate the driver that he has arrived in position. The bumpers are separately installed outside onto the side walls of the pit. No loading platform should be without them. They absorb the forces that are generated each time a vehicle docks, thereby ensuring that the dock shelter and platform do not get damaged in the process.

- Made of anti-friction full rubber
- Available in various sizes
- Front-top hot dip galvanized protection plate
- Spring balanced up-down Floating Bumpers move 150 mm up and 90 mm down and rest at dock rim

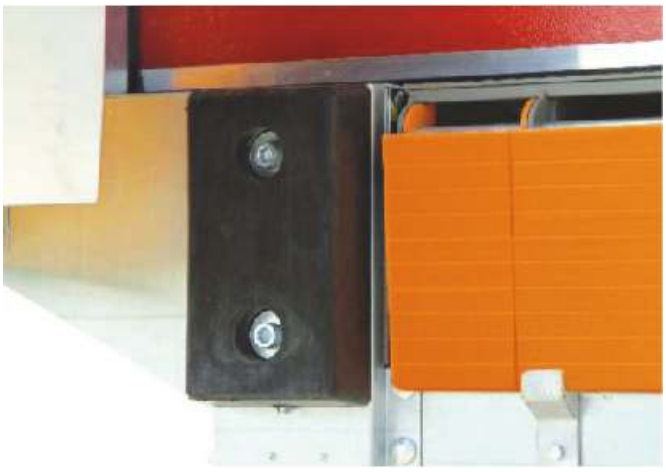


Super bumpers fully respond to the necessity of safeguarding the front facade of the building.

- Made of anti-friction full and hard rubber
- 150 mm width, 400 mm height, 80 mm depth with heavy steel support (total depth including steel support, 130 mm)

Mega bumpers larger in dimensions safeguards the front facade of the building.

- Made of anti-friction full and hard rubber
- 250 mm width, 450 mm height, 100 mm depth with heavy steel support (total depth including steel support, 150 mm)





Bumpers with Steel Front Plate

Where greater wear and tear is expected, protective plates on the front of the bumpers are recommended.

- Hot deep galvanized steel plate for protection front and top loads
- 10 mm thickness for Super bumpers
- 15 mm thickness for Mega bumpers

Accessories for Safe Docking





Wheel-lok Vehicle Restraint System

The most sophisticated electrical hydraulic technology, operated via a microprocessor and is controlled using a button on the control panel in the building. An integral sensor in the wheel block accurately determines the position of the rearmost wheel, where upon the wheel block is automatically extended and the wheel is restrained. The restraint process occurs in approx. 30 seconds regardless of the size and position of the wheel.

- The wheel block applies constant pressure against the wheel to prevent the vehicle from creeping forwards
- No damage to mudguards or side skirts due to the low height of the wheel block. Warning lights are in constant operation while the wheel block is being positioned and an external red/green traffic light ensures clear communication
- Emergency release possible at all times



Vehicle Restraint Safety Wedge

The wedge is put under a vehicle wheel, as to restrain its moving, which increases the safety-on-work.

- Heavy duty type, with rudder and positioning wheels
- Construction heavy duty rigid plastic component sandwich side covered with 8 mm thick zinced steel plates, blocked by passing through tyres

Rotating Hook Automatic Vehicle Restraint System

Integrated vehicle restraints improve dock safety by securing the truck during the loading operation by engaging the trailers rear impact guard with a 32,000 lb. restraining force.

- Large rotating hook design helps prevent unscheduled trailer departure, creep and tip over
- Upward biased hook automatically adjusts to trailer movement keeping constant contact with rear impact guard
- Improved direct drive system eliminates clutch, chain and brake. Exclusive water tight hook position sensor eliminates limit switches
- Inside and outside LED communication lights communicate all operational modes to the truck driver as well as the dock attendant





Anti-Crushing Dissuaders

The anti-crushing safety dissuaders are flexible but resilient volumes applied on the outer wall of the loading bay, filling the space between the wall of the building and the pillars of the containers of the vehicles. This dramatically reduces the risk of an operator slipping into the space and subject to crushing between the vehicle and the wall in the fatal case of a sudden manouevre of the vehicle. The vehicle while docking can compress the dissuader without damaging it, but when it goes away it will return to its original size.

Dock Levelers for Special requirements



Mechanical Dock Levelers

Mechanical Dock Levelers are designed with the industry's most reliable holdown and a full yielding walkout lip extension for the smoothest and easiest operating mechanical Dock Leveler.

Standard features include working range side guards, retracting solid steel deck legs, night security lock, end-loading control and integral maintenance support.

Air-Powered Dock Levelers

The self-contained design of Air powered Dock Levelers provides easy handling & installation with a fully protected bag and a lift system that does not contact the floor of the pit.

Standard features include full operating range telescoping side guards, night security locks, integral lip and deck maintenance supports, safety black & yellow side skirt labels, and two heavy-duty dock bumpers.



Dock House



Dock House is a self-contained loading bay which is installed in front of the building providing energy efficiency, contributing to hygiene, safety and is space-saving. This means that the actual building does not have to undergo any expensive structural alterations. Dock Houses do not take up valuable internal space of the building and forms a complete loading house that is ideal for ambient and temperature controlled buildings. The design and layout can be adapted to suit the requirements of the site.

Dock Houses are used in the following situations:

- When the manoeuvring space for trucks is insufficient to allow them to dock at a 90° angle
- When extending or converting existing buildings
- When Dock Levelers have to be positioned outside the building in order to avoid thermal bridges

Dock House is made up of following components:

- Dock Leveler
- Dock Shelter
- Sectional Overhead Door
- Roof and Side comprise a sturdy steel frame construction which is clad with double-skinned steel puf panels

Sectional Overhead Doors - Quality Engineered





Porto Sectional Overhead Doors are the ideal solution for all industrial and logistic needs. The design and different solutions offered ensure the door to be aesthetically pleasing and perfectly suited in any architectural environment - from modern and traditional industrial buildings to fine commercial buildings. These doors are built to ensure the highest ease and flexibility of use which, in turn, ensures a quick, hassle free and accurate replacement of old doors. Their compact size ensures more available space both inside and outside the premises. Depending on the structure of the building and the requirement a choice can be made from a standard lift, vertical lift, horizontal lift, low headroom or inclined lift.

Porto Sectional Doors come in a wide variety of designs with various features to suit your individual requirement, all guarantee

- Reliable and low-noise operation
- Extreme robustness
- Safe operation in compliance with safety requirements
- Design-oriented surfaces and optimum light solutions

Porto

Double-skinned Steel Sectional Doors

Panel

The Panels are sturdy and consist of double-walled steel with a polyurethane hard foam core which is foamed CFC-free. The standard thickness of the sections and therefore of the entire door leaf is 40 mm, which provides superb heat insulation and soundproofing. The double-skinned 80 mm thick panels suitable for refrigerated warehouses and storage facilities in the food sector is available on special request.

The door panels are available in two heights 500 mm and 610 mm.

They are connected with each other by galvanised steel hinges. This guarantees safety and durability.



Technical Characteristics

- Density : ± 45 kg/m³
- Adhesive strength : ± 0.10 MPa
- Compressive strength : ± 0.13 MPa
- Compressive Modulus of Elasticity : ± 2.68 MPa
- Thermal conductivity : ± 0.14 W/mK
- Heat transfer co-efficient:
  - Panel 610 mm with seal : U= 0.76 W/m²K
  - Panel 500 mm with seal : U= 0.80 W/m²K
  - Panel 610 / 500 mm without seal : U= 0.53 W/m²K
- All panels are lined internally and externally with a protective PVC film

Colours & Finishes Offered

Standard External Microrib



WHITE
similar to RAL 9010



GREY
similar to RAL 9006



BLUE
similar to RAL 5010

Standard Internal Stucco



WHITE
similar to RAL 9010

Optional External Microrib Internal Stucco

Any RAL colour on request to match the overall appearance of building





Our Sectional Door design is optimal for customers who need robust, well-insulated and space-saving doors with safety and offer amazing advantages.

Compact Design

Sectional Overhead Door side runners move vertically along the wall and parallel to the ceiling, ensuring optimal use of the available interior space.

Environmental Control

Their heat insulation and soundproofing ensured by insulated panels, improves internal living conditions and energy consumption.

Safe and Secure

First class materials, precise processing and continuous quality checks guarantee perfect safety, reliability and long service life. All our doors comply with safety requirement of European standards EN 13241-1.

Glazing for Light & Aesthetical Look

The panels can also be designed to be combined with thermally insulated panels and aluminium panels featuring transparent acrylic, grilled or meshed windows. This design allows natural light to go through large clear area.

Installation Friendly

Our door installation is fast and simple due to intelligent product optimization without having to compromise on proven quality. Numerous prefabricated parts prevents faults, reduce the installation time and increase efficiency.

MaxVista



MaxVista is a Sectional Overhead Door designed to be combined with thermally insulated panels and extruded aluminium panels, shaped for inserting transparent acrylic, grilled or meshed windows. This combination gives the door a very distinctive look, which enhances the face of both commercial and industrial buildings, where MaxVista blends in perfectly as a practical and decorative feature.

It makes the environment light and pleasant to work in as it allows natural light to go through the large clear area. The aluminium profiles and the glass window sections are highly resistant to the elements and ensure a longer life of the door.



Quality that ensures Safe Door

Silent Door Travel

Robust roller holders made of galvanised steel with adjustable nylon rollers on ball-bearings guarantee a long service life and silent door travel.

Intermediate Hinge

The stable centre hinge is made of galvanised steel connect the individual door panels precisely.

Finger Trap Protection

The horizontal joints between the door panels are designed in such a way that it avoids the risk of fingers being pinched, if a hand if rested on the moving door panel, both on the inside and outside of the door leaf.

Side Trap Guard

The side frames are completely closed from top and bottom and hence injuries are not possible due to intervention from the side.

Cables Guided Internally

The carrying cables are guided on the inside between the door leaf and frame. The risk of injuries are excluded as there are no protruding components.

Seals

The seals are made of special anti-aging EPDM are located in the following hinged points;

- Central seals between the individual panels across the entire width
- Upper doorstop finned seal fitted to seal the door near the lintel
- Vertical seals fitted to the side uprights
- Bottom Seal with rubber profile helps even to compensate for any unevenness in the floor

They ensure total insulation thus avoiding the ingress of air, dust and water.



Safety Features

In Accordance with European Standard 13241-1 CE



Safe Door Guidance

The rollers are guided precisely in a safety track ensures derailment guarded operation of the door installation.

Torsion Springs

The door leaf weight counterbalance on Sectional Doors in standard dimensions is designed with torsion spring assembly. The steel torsion springs have been designed to ensure a standard life from 10,000 to 1,00,000 operations cycles.

Safety Device Against Spring Breakage

This special device immediately stops the door panel from closing in the event of balancing spring breakage.

Safety Device Against Cable Breakage

This special device immediately stops the door panel from moving in case of breakage of the return cables.

Closing Edge Safety Device

A pair of Photocells ensures safe monitoring of the closing edge of the door, stop and reverse if any obstruction is detected before they come into contact with the door.

Special Requirements

Over Dimensional Sectional Door for large openings with Safety



We fabricate precisely to your application and building openings that implies to our selection of perfect design, colour, size and maximum stability. Our Sectional Doors fit openings to a maximum width of 15,000 mm and height of 10,000 mm. Future developments and continuous improvement of existing systems are our standard procedure.

Special Requirements

Side Door or Pedestrian Door



For additional easy access for people and small transport vehicles, a separate threshold-free side door or an integrated pedestrian door can provide access even when the Sectional Door is closed. This also saves the heat loss from the building by decrease in the number of times the door is operated making it energy efficient.

If there is enough space, a side door separate from the door installation is a good choice. We offer side doors on request in the same design as the door installation. A pedestrian door integrated in the Sectional Door is always an expedient supplement if there are no space for side door installation. All pedestrian doors open outwards for safety reasons. A special switch ensures on electrically-driven Sectional Doors that door can only be moved when the pedestrian door is closed.



Types of Sectional Door



Manual Push-Pull Sectional Doors

This type of Sectional Doors are operated by pulling or pushing the doors manually. Counter balancing of the door weight is done using torsion springs within strong suspension shaft to reduce deflection. Due to manual exertion, these doors are ideal for very small size upto width 3000 mm x height 3000 mm.

Chain-Hoist Operated Sectional Doors

This type of Sectional Doors are operated manually by pulling an endless chain. They are relatively easy to operate as compared to Push-Pull type sectional doors as they roll up and down smoothly. Counter balancing of the door weight is done using torsion springs within strong suspension shaft to reduce deflection. The hoist is installed directly in the shaft to enable the manual lifting of the door. These doors are ideal for size upto width 8000 mm x height 6000 mm.





Electrically Operated Sectional Doors

This type of Sectional Doors are operated by using a range of electrical actuators made up of self-braking worm gear with hollow shaft and integrated limit switches with overload protection. During downtime, the option for manual operation is also available.

Optional Accessories

All Sectional Doors are equipped with a set of accessories that provide them with all necessary safety, aesthetic elements and robustness for perfect operation. There are, however, a series of optional accessories that may be added to a final product according to the wishes of the customers. Each accessory is subjected to rigorous tests before being selected and included in the production process, to ensure the highest quality of the product that will fulfil our customer needs.



Portholes

Rectangular window in acrylic material in black profile, Dimension width x height: 636 mm x 333 mm



Ventilation Grill

These avoid condensation and vent fumes to the outside. Available in white or black plastic, ABS material. Dimension width x height: 344 mm x 138 mm



Flashing Light

Visually signifies door operation. Flashing light serves safety requirements. They give visual signals for opening and closing impulse.

Plastic housing IP 65

Interlocking System - Dock Leveler & Sectional Door

This prevents operation of Dock Leveler when the Sectional Door is closed or only partially open and also prevents door from closing while the Dock Leveler is still in a working position.

Interlocking System Between Two Doors

Developed for climate control or safety, paired doors interlock with control transfer function. If door A is open, door B cannot be opened. If door B is open, door A cannot be opened.



Infra-red Photocells

The panel down movement is automatically stopped or reversed if the infrared beam is broken. We distinguish between two functions on the photocells

a) Open : The door opens when the photocell beam is interrupted.
b) Safety : The door remains open as long as the photocell beam is interrupted.

Mounted on walls or columns in the vicinity of a door.





Single Station Push Button

Plastic housing IP 65

Mushroom Top Push Button

With large operating surface Plastic housing IP 65



Three Station Push Button

"Open/Emergency-Off/Close"
Plastic housing IP 65

Types of Installation



Vertical Lift (VL Elevation)



Normal Lift (NL Elevation)

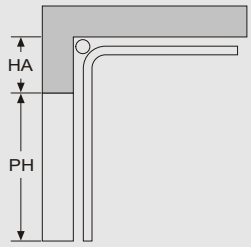


High Lift (HL Elevation)

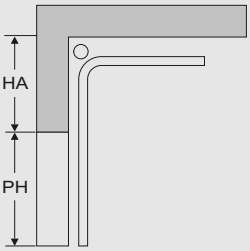
Sliding System Vertical Section Drawings: Types of Installation

Minimum lintel size - 200 mm

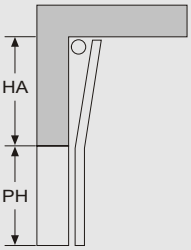
NL = normal lift



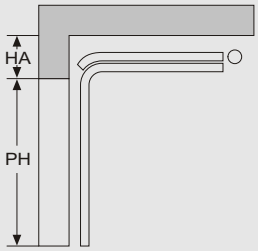
HL = high lift



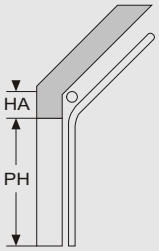
VL = vertical lift



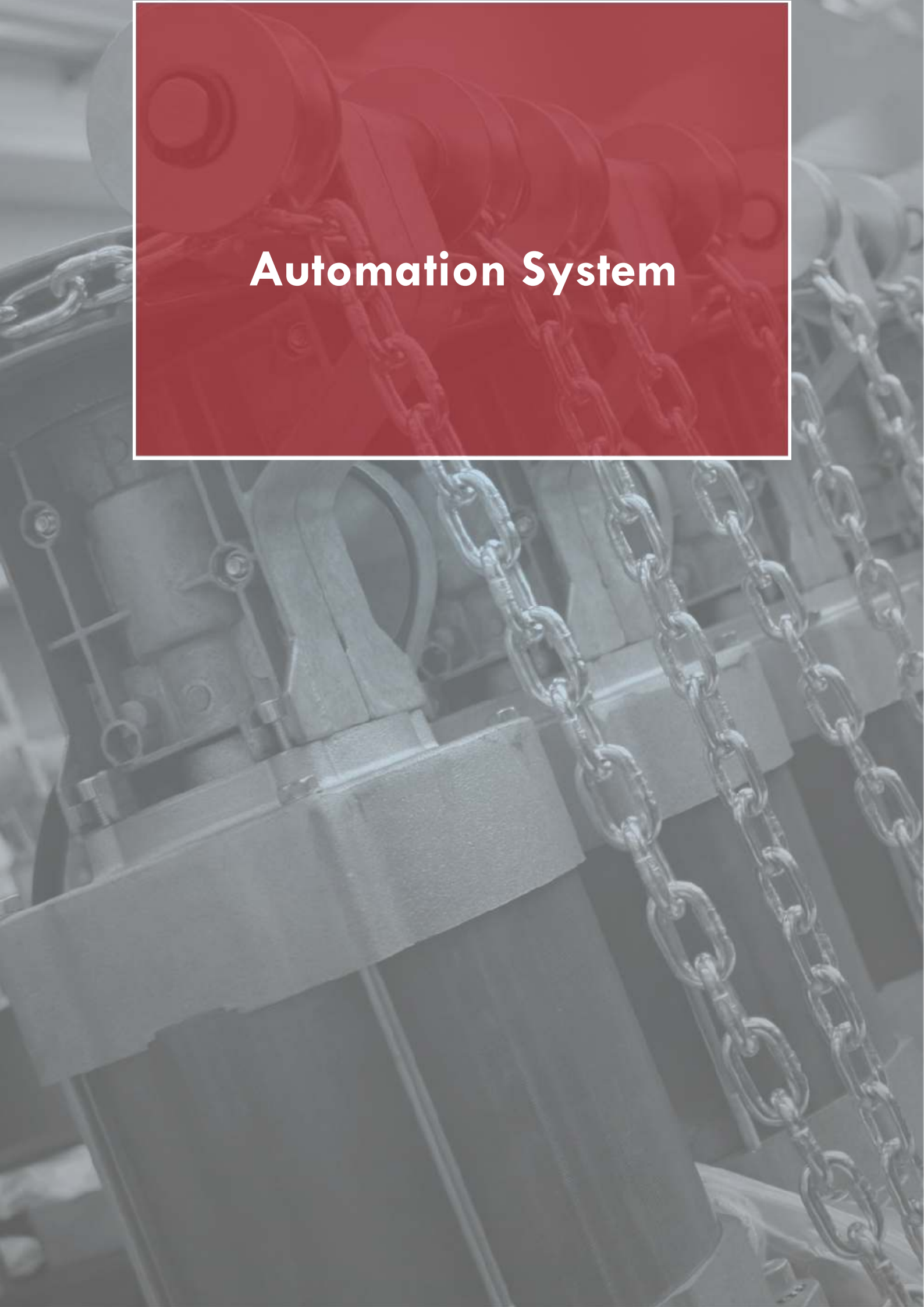
LHR = low head room lift



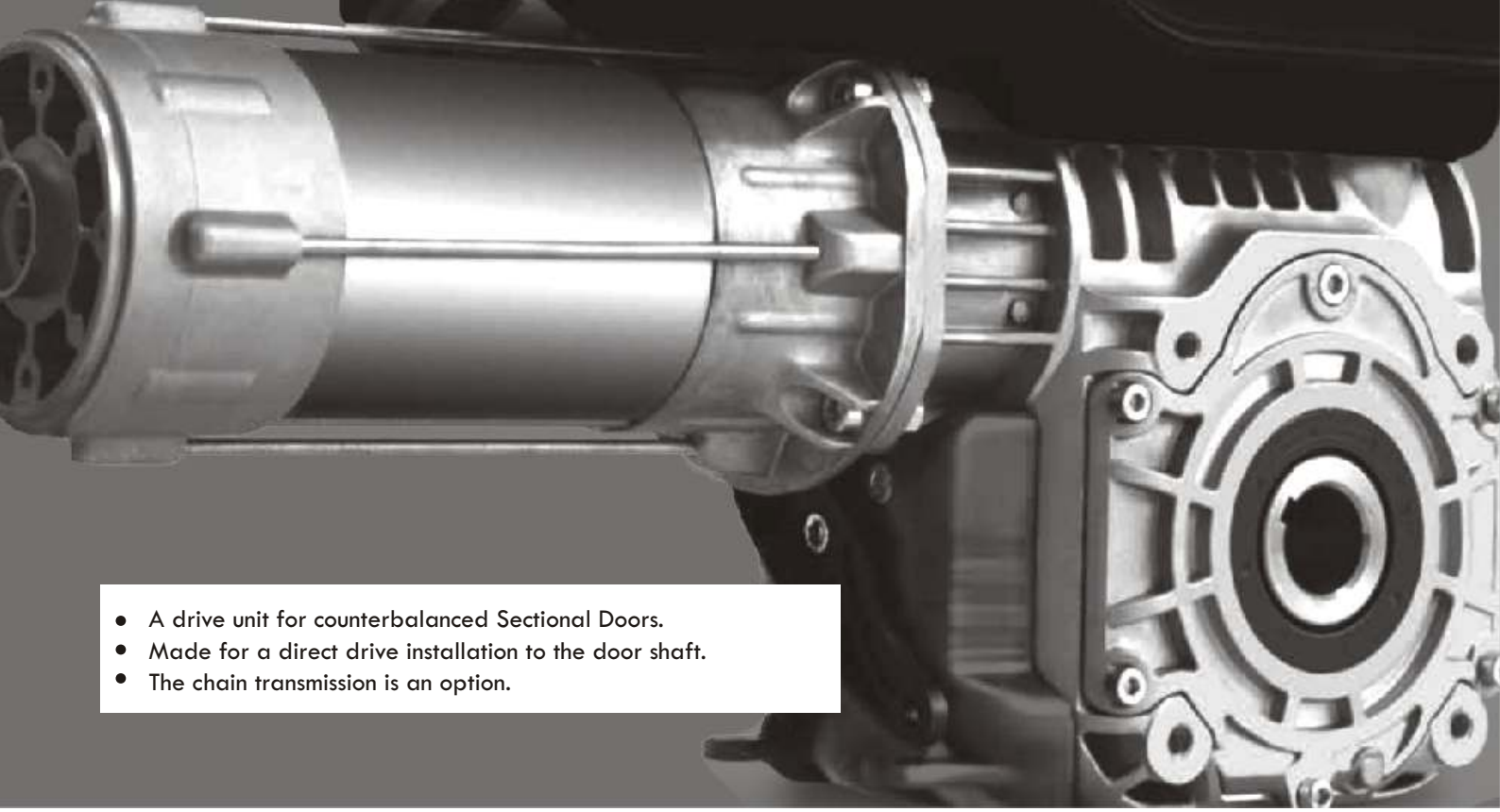
IL = inclined lift



Elevation Details	PL (mm)		PH (mm)		HA (mm)	
	Min.	Max.	Min.	Max.	Min.	Max.
Normal (NL)	1800	8000	1780	5570	420	
				6500	510	
				8000	650	
High (HL)	1800	8000	1780	5500	960	4000
				8000		
Vertical (VL)	1800	6500	1780	5000	PH+300	
		8000		8000	PH+650	
Low Head Room (LHR)	1800	5000	1780	3500	200	
Inclined Lift (IL)	1800	6000	1780	5000	510	
		8000		8000		



Automation System



- A drive unit for counterbalanced Sectional Doors.
- Made for a direct drive installation to the door shaft.
- The chain transmission is an option.

Drive Unit is made up of

- A worm gear with hollow shaft
- An electric motor
- Integrated limit switches
- An emergency manual operation

The self-braking worm gear makes the drive irreversible. No need for an electric brake.

Additional Details

European Directives

Drive Unit is made in accordance with product standards EN 13241-1 Doors and EN 12453 Safety for power operated doors.

Overload Safety Factor

Motor overload safety factor = 4 x the the nominal motor amperage because the starting current of the drive can reach these levels for short periods.

Manual Operation

In installations of drive with hand crank or hand chain override, the door and self braking gear construction remain inter-connected. No danger of the door crashing down if a spring breaks.

The counter-balancing system should be inspected at least once a year.

Duty Rating

The relation between the weight of the door and the number of movements per hour allows to increase the number of movements by reducing the door's weight.

Output Speed

The maximum admissible speed depends on the construction and type of door. The admissible closing speed shall be tuned so that the operating forces must comply with EN 12453.

Holding Torque

The drive unit is capable of holding the weight of the door when the spring breaks preventing the door leaves from crashing down.

The holding capability is the admissible load bearing of the worm gear construction.

Drive Units

Motors with integrated safety brake for industrial and commercial Sectional Overhead Doors.



Override options

Emergency manual operation types

- Hand Chain Override Standard
- Hand Crank Optional

Limit switch options

Limit switch types

- Mechanical Limits
- Mechanical Limits (2 cams)

Technical details

Description	GAPL70MCH	GAPL90TCM
Torque (Nm)	70	90
Output Speed (Min^{-1})	21	21
Hollow shaft (mm)	25.4	25.4
Holding Torque (Nm)	450	450
Power (kW)	0.50	0.80
Power Supply (V~)	230	400
Frequency (Hz)	50	50
Amperage (A)	2.20	2.20
Max Cycles Per Hour	12	20
Limit Switch range	18	18
Protection Rate (IP)	54	54

GAPL70MCH motor is the standard drive

Higher range available on request

Control Units

QC0P

Multifunction control unit with radio receiver



QC0P is a multifunction control unit in an IP54 (IP65 upon request) ABS box, which enables the remote control of door motors up to 500 W. It allows the control of the different safety devices installed.

Features

- Operating protocol: momentary, maintained
- Remote control
- Safety beams (active also for opening)
- Safety edges (resistive type with test feature)
- Pause timer and automatic closing
- Flashing light/courtesy light
- Adjustable working time, from 3 seconds to 4 minutes

Technical Specifications

Description	QC0P
Power supply (v~)	230 - 50 Hz (±10%)
Frequency (mhz)	433.92 MHz
Power supply External accessories	24 V~ - 50 Hz/Max 140 mA
Number of radio codes	63
Protection rate	IP 44
Motor power	max 230 V~ - 500 W
Working temperature	-10°C /+60°C
Working time	default = 30 sec
Dimensions (mm)	145 x 145 x 59
Weight (g)	340

Control Units

QC3P

Multifunction control unit with integrated push-button



QC3P is a control unit in an IP44 (IP65 upon request) ABS box, which enables the remote control of Sectional Overhead Door motors up to 2000 W.

Features

- OPEN / CLOSE / STOP membrane buttons
- Inter-locked contactors
- Plug-in connection cables
- Available with optional ready-wired 16 A CEE plug (it replaces the main switch according to EN 12453)
- Operating modes:
  - Hold to CLOSE / OPEN
  - Hold to CLOSE / AUTOMATIC UP
- Additional connections for: Push-button and Safety beams

Technical Specifications

Description	QC3P
Power supply (v~)	230/400 V~ (3Ø) - 50 Hz (±10%)
Power supply External accessories	24 V~ - 50 Hz/Max 70 mA
Motor power	max 400 V~ - 2000 W
Protection fuse	315 mA
Protection rate	IP 44
Working temperature	-10°C /+60°C
Dimensions (mm)	145 x 145 x 59
Weight (g)	1000

QCO3+

Multifunction electromechanical control unit



QC03+ is a multifunction electromechanical control unit in an IP44 ABS box which enables the radio control of three phase Sectional Overhead Door motors up to 2000 W. It features 2 mini-inverters which assure the highest reliability on powerful motors. It allows the control of the different safety devices installed.

Features

- Operating logic: momentary, dead-man (only for closing)
- Remote control
- Safety beams (2 inputs)
- Safety edges (test feature)
- Pause timer and automatic closing
- Flashing light/courtesy light
- Adjustable working time, from 3 seconds to 4 minutes

Technical Specifications

Description	QC03+
Power supply (v~)	400 V~ - 50 Hz (±10%)
Power supply External accessories	24 V~ - 50 Hz/Max 250 mA
Motor power	max 400 V~ - 2000 W
Protection fuse	315 mA
Protection rate	IP 44
Working temperature	-10°C /+60°C
Dimensions (mm)	225 x 190 x 88
Weight (g)	2450

Dock Shelters & Seals



Cutting off the inside environment from the outside, it is not only useful to the human working conditions but also for avoiding useless wastes, therefore resulting into energy saving, ecological safeguard, protection of perishable goods and greater working efficiency.

Gandhi Automations offers a complete range of Dock Shelters available for every kind of use and environment; to protect the internal heated environments from the outside cold conditions or the chilled ones from outside heat and also from rain, wind, snow & dust.

Strong and of good quality, the different models are designed to resist to the docking of the vehicles, reducing their impact in different ways.

Better Dock Safety

Dock Shelters seal the gap between the building and the vehicle in such a way that when the Sectional Overhead Door is opened goods and personnel are protected against the harsh weather conditions outside.

Dock Shelters provide a seal between the internal and external environments thus assisting in reduction of energy consumption. The savings in energy costs is considerable.



Dock Level



Ground Level

Retractable Dock Shelters

With a Dock Shelter installed, the truck and dock form one closed entity, shutting out all risks form outside. Both for new and existing buildings we can always realize an optimum sealing, when necessary with a customized dock shelter.

The Retractable Dock Shelter is suitable for all vehicle sizes and has flexible design. Retractable steel framework is supplied with hinged arms to provide a shelter which if struck by misaligned vehicle easily retracts behind the bumperface without any damage to the shelters.

The Retractable Dock Shelter is the standard solution for energy-saving oriented operators. The vehicle reverses into the dock shelter which seals it off with flexible side and top curtains, giving weather protection during the loading and unloading process. The result is an improved working environment and goods protection. The curtain material has a very high wear and tear resistance.

The trailer backs into the Dock Shelter, with the driver able to centralize upon the clearly visible chevrons on the side curtains. Driving approx. 500 mm into the shelter, the curtain forms an excellent seal around the trailer body. The curtains are pressed out of the door opening creating a perfect entrance for fork lifts into the trailer. The standard width of the shelters is 3500 mm. Over the years this has proven to be the best dimension, giving the best result over a very long period.

Frame is made from articulated steel zinced retractable structure with anti-pinch retraction arms. Specially developed corner pieces of rugged construction give an excellent rigidity to an assembled framework. The rare and front frame is connected by arms with spring integrated at both ends.

Retracting and self-returning system operating on swinging arms, connecting the front to the rear part. They allow the Dock Shelter to retract is case of wrong manoeuvre of the lorry, enabling the automatic return in position and damage will be minimized.

Side and top leaf of standard yellow PVC, bright at the inside with the day light, inset into the throats of the perimeter frame.

Top horizontal leaf 1000 mm height and side horizontal leaves 630 mm width, thickness 3 mm double waving hard armored, for the best adherence to the vehicle sides.



Cushion Dock Shelters

Cushion Dock Shelters are best suitable for loading dock where many vehicles of approximately the same height and width load & unload and where environmental control is important. It provides a virtually air tight seal between the vehicle and building. Even the gaps between the trailer door and bodies are sealed.

Only a cushion shelter design offers you this advantage. This innovative cushion shelter has been specially designed for warehouse with the highest demands regarding the inside temperature, like cold stores and freezer stores.

Since the foam cushion slightly reduces the width of the available door opening, non-palletised goods, are recommended to avoid side cushion damage during loading and loading.

Construction

The three cushions are made of elastic polyurethane foam, covered with PVC coated polyester fabric, supporting the vehicle pressures and perfectly sealing the three sides, including the space between opened rear doors and sides of the vehicle.

The covers are made of high quality vinyl, or of extremely highly wear resistant polyurethane. The vinyl material strengthened with a fabric insertion is suitable for use in moderate climates. (Vinyl loses its flexibility at temperature below -10°C and it can break down at temperature below -25°C). Polyurethane is eminently suitable for situations of extremely frequent use and /or temperatures below -10°C.

The two vertical cushions have continuous overlapped anti-friction limpets allowing for the up and down heavy friction of the vehicle on its suspensions, during the loading.

The foam cushion of the seal has a recovery rate of 98% and remains flexible in temperatures down to -40°C. It is resistant to most solvents, chemicals, moulds and fungi. The seal covers employ ventilation exhausts vents to ensure rapid compression and decompression without problems. Bold guidance markings at the front of the seal side cushions are useful point of orientation when reversing onto the dock seal to reinforce this correct alignment we strongly advise the use of wheel guide.



Ground Level



Dock Level

Inflatable Dock Shelters

Inflatable Dock Shelters combines an optimum seal and a virtually airtight contact with the vehicles. Ideal for temperature controlled warehouses to seal against summer heat, winter cold, draught, dust and insects. The result is substantial savings on energy, improved working environment, production and safety. Not only do the inflatable side and top seals provide the best possible closure, they also accommodate a wider range of vehicle sizes varying up to 1200 mm in height and 700 mm in width. This feature means that trucks of varying dimensions can still be sealed effectively.

Flexibility Combined with Optimum Closure

The Inflatable Dock Shelter is made of polyester fabric PVC covered, a material resisting to hot temperatures and bad weather conditions. The frame is exceedingly robust and consist of 40 mm insulated panels. The standard cladding color is 'white or grey'. The pneumatic Dock Shelter is to be inflated only when the vehicle is in loading position. The fan can be switched off at the end of the operation and the cushion will rapidly retract to let the vehicle go. Inflatable Dock Shelters provide the most versatile seal available to service the widest variety of truck and trailer configurations.

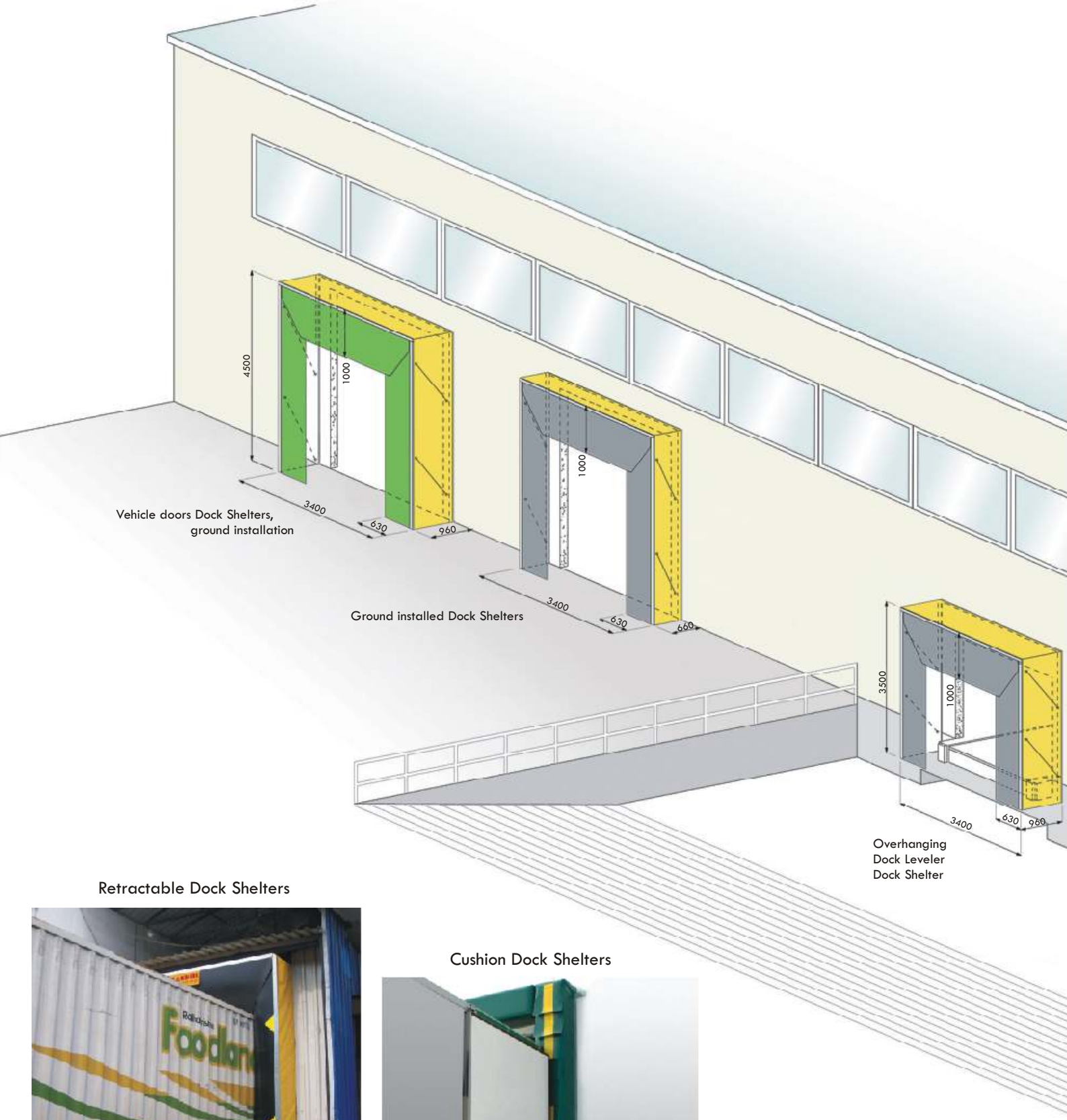
Contrary to other types of Dock Shelters, the vehicle does not push towards the shelter. Instead the shelter is inflated around the docked vehicle providing complete seaming.

The three cushions inflate under 230 V 50 Hz single phase 0.55 kW fan air pressure until light compression against roof and sides of the lorry. The cushions retract rapidly to their rest position when the fan is switched off for allowing the departing lorry to be immediately free from cushion pression. The inflatable side and top seals provide a virtually airtight contact with the vehicle.



*For Inflatable type of Dock Shelters it is strongly recommended to install / fix ground wheel-guides for perfect docking of trucks. This prevents damage to the side panels of Dock Shelters.

Dock Shelters & Seals



Retractable Dock Shelters

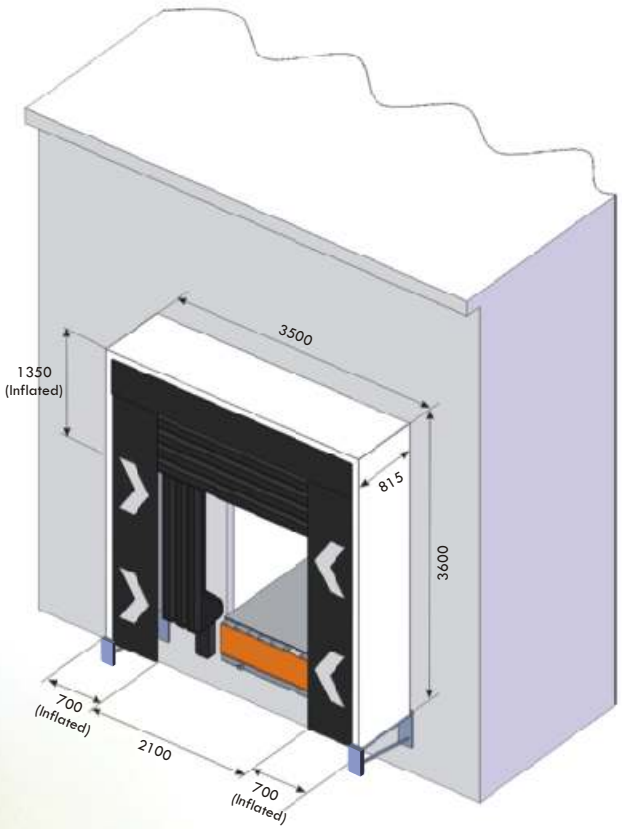


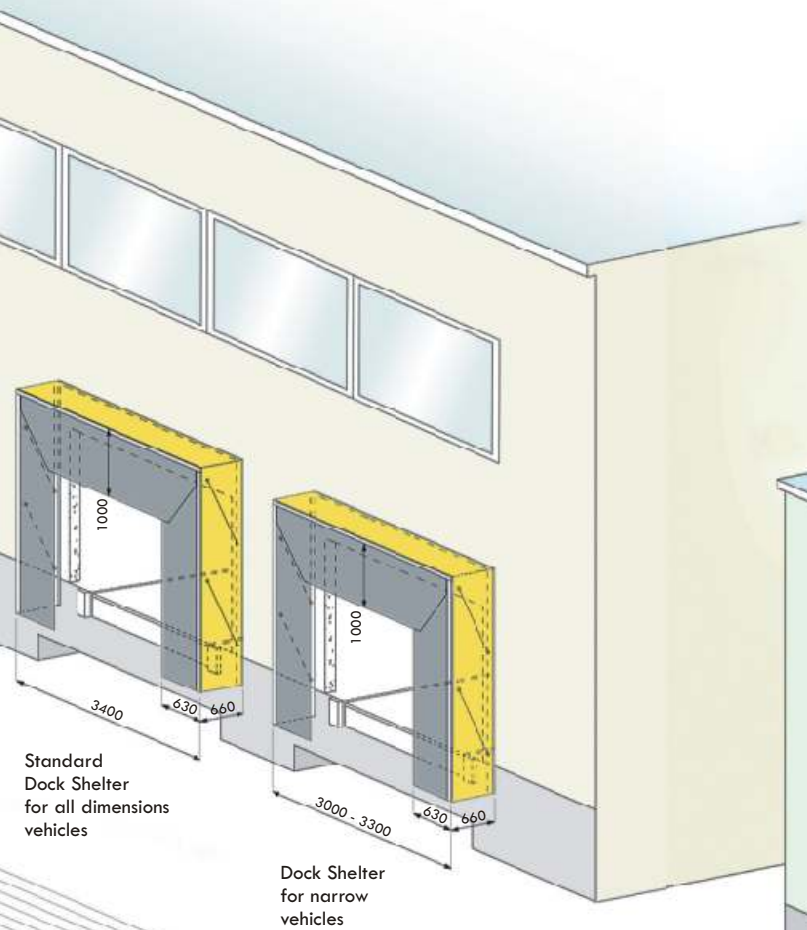
Cushion Dock Shelters





Inflatable Dock Shelters





Standard
Dock Shelter
for all dimensions
vehicles

Dock Shelter
for narrow
vehicles

